Sleep Study on the Job

An investigation into sleep qualities and occupation.

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In this report, we aim to derive potential connections between sleep behaviors, and occupation. We will be working with a dataset containing a mixture of occupations, and utilizing sleep quality, sleep duration, and stress levels, to measure correlations.

Importance and Context

We plan to study an answer to the question: "What is the difference in sleep quality between different occupations". This question is interesting to us, as we believe it is a common belief that some occupations demand more hours, or are more stressful than others may be. We wonder if this thought is based in fact, or if people, when studied, claim and behave differently in their twilight hours. There are also related questions we could seek to answer, such as "Does occupation have a notable relation to stress level?" and "Does Occupation have correlation to physical activity level?". The question of whether occupation affects sleep has been studied before, with a mixture of results depending on the study, and individuals polled. Some look at the prevalence of short sleep in certain career fields (Oxford Academic). While others look at associations between working hours and sleep (Taylor & Francis). And some study occupational stress and sleep (NIH). We seek to evaluate a person's claim of sleep quality first and foremost alongside their sleep duration to get a strong measurement of their occupations' possible connection.

To answer this question, we have a dataset containing occupations, sleep durations, quality, stress level, physical activity level, and even steps. We can derive information from this, by comparing each occupation and finding unique takeaways from this dataset involving sleep duration/quality and career field specifically. We can use linear regression models to find correlations between sleep quality/duration and occupation in our studies, looking in particular for p-values of notable significance. Scatter plots could be useful as well to show trends and potential clustering towards, or away from high sleep quality/duration when considering different occupations. We are excited to immerse ourselves further into this data, and derive potentially notable insights.

Data and Methodology

In our research, we utilized a data set containing sleep and health/lifestyle data with 374 individual unique Person IDs. (add information on Outcome variable, observation types,)

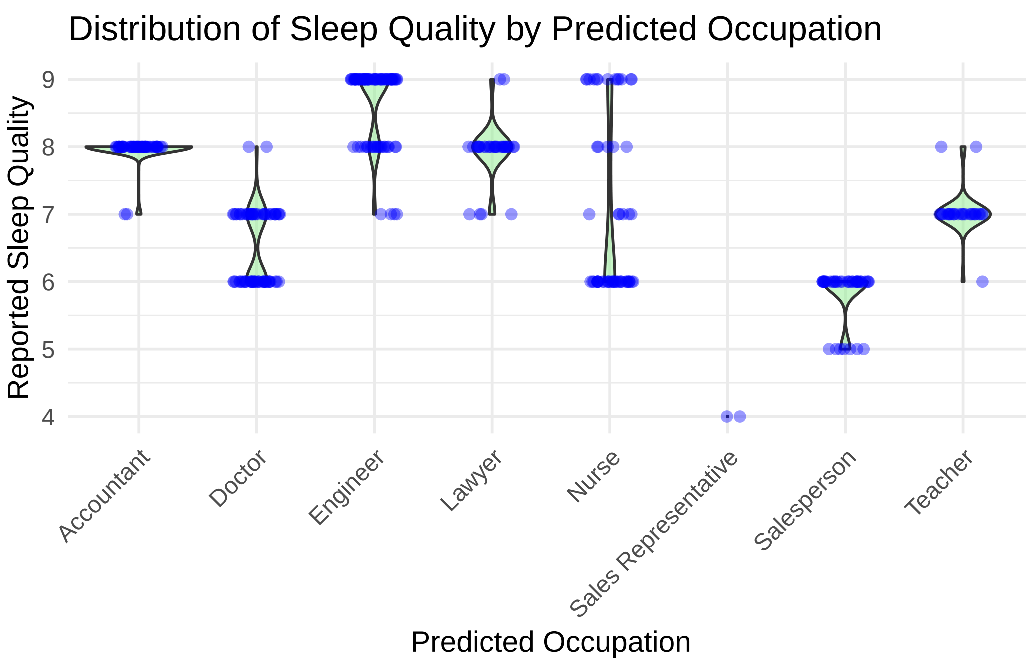
Filtered Data Focusing on Most Telling Variables:

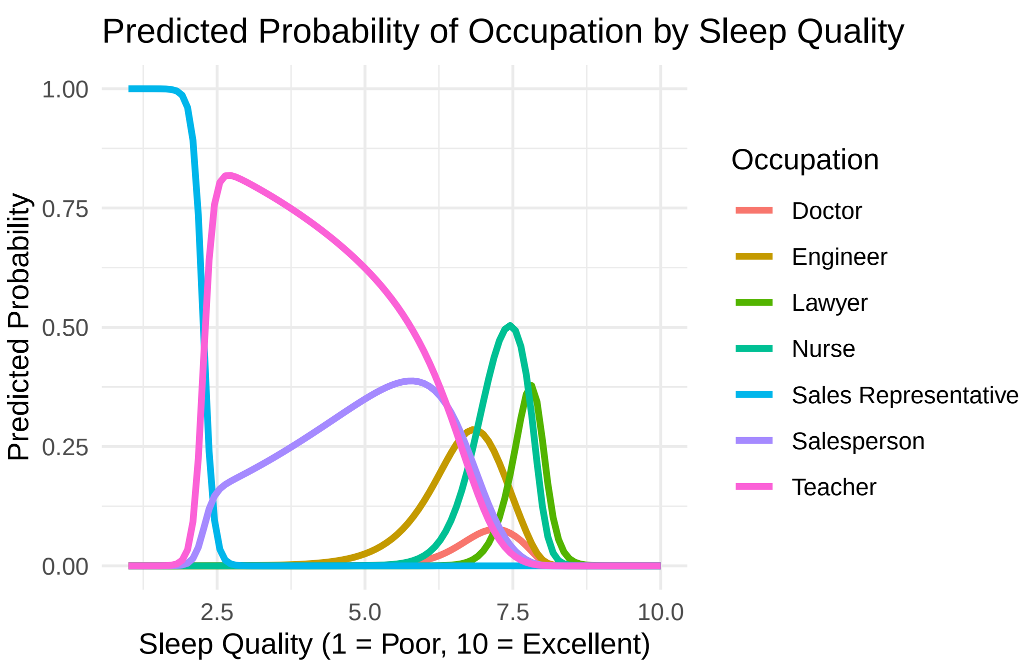
```
Call:
multinom(formula = Occupation ~ Sleep.Duration + Quality.of.Sleep +
    Stress.Level, data = filtered_data)

Coefficients:
                     (Intercept) Sleep.Duration Quality.of.Sleep Stress.Level
Doctor                -76.049759       20.98580       -11.518211    2.6307859
Engineer              -37.245006       19.19423       -12.653930   -0.4832950
Lawyer                -63.402905       15.68993        -7.431006    1.8425532
Nurse                 -24.575136       15.14807       -10.580911   -0.1167972
Sales Representative    7.640907       12.31122       -26.566682    7.8627480
Salesperson             9.851511       14.93626       -14.250728   -1.6100903
Teacher                38.367501       12.76148       -14.669175   -3.5304241

Std. Errors:
                     (Intercept) Sleep.Duration Quality.of.Sleep Stress.Level
Doctor                14.4032270       3.053222         2.302738    0.7445940
Engineer              12.1714155       3.013075         2.335805    0.6263973
Lawyer                12.3521966       2.991638         2.243352    0.5850389
Nurse                 10.1477268       2.935857         2.247683    0.4906206
Sales Representative   0.4463054       2.635039         1.790183    3.5675582
Salesperson           14.1861641       3.125717         2.327490    0.6426961
Teacher               13.7885672       3.147872         2.356832    0.6695928

Residual Deviance: 808.2145
AIC: 864.2145
```

Filtered Data of Only Sleep Quality:

```
Call:
multinom(formula = Occupation ~ Quality.of.Sleep, data = sleep_data)

Coefficients:
                     (Intercept) Quality.of.Sleep
Doctor                10.2906286     -1.327624158
```

```
Engineer              -5.7617832    0.769908796
Lawyer                 0.2189951    0.002548614
Manager                3.3511922   -0.934527549
Nurse                  4.8782140   -0.549760211
Sales Representative  51.4615284  -11.041896782
Salesperson           16.5407701   -2.447179109
Scientist             22.9237972   -3.971136249
Software Engineer      8.7001914   -1.523271183
Teacher                7.1488588   -0.950334347

Std. Errors:
                     (Intercept) Quality.of.Sleep
Doctor                  1.772686        0.2366432
Engineer                2.106038        0.2569680
Lawyer                  1.888777        0.2376973
Manager                 7.393940        1.0434658
Nurse                   1.680487        0.2156816
Sales Representative  124.078327       31.0060686
Salesperson             2.562490        0.3864753
Scientist               4.241898        0.7531106
Software Engineer       4.242499        0.6342044
Teacher                 1.871204        0.2481227

Residual Deviance: 1283.703
AIC: 1323.703
```


Distribution of Sleep Quality by Predicted Occupation
Reported Sleep Quality
9
8
7
6
5
4
Accountant
Doctor
Engineer
Lawyer
Nurse
Sales Representative
Salesperson
Teacher
Predicted Occupation


Predicted Probability of Occupation by Sleep Quality
Predicted Probability
1.00
0.75
0.50
0.25
0.00
2.5
5.0
7.5
10.0
Sleep Quality (1 = Poor, 10 = Excellent)
Occupation
Doctor
Engineer
Lawyer
Nurse
Sales Representative
Salesperson
Teacher

Results

Discussion

Final Report

Your final report should document your analysis, communicating your findings in a way that is technically precise, clear, and persuasive.

Page limits:

- Main report: 4 pages
- Appendix: 1 additional page

You must meet these page limits using standard `pdf_document` output in a `.Rmd` file, or in a `documentclass: scrartcl` in a `.qmd` file. (This assignment prompt has been generated as a `scrartcl`).

The one-page appendix is intended for extra information that will help your instructor assess your model building process. Please include the following elements:

1. **A Link to your Data Source.** If you used specialized code to access your data, please include that here. Please make sure your instructor has the ability to access the data.
2. **A List of Model Specifications you Tried.** We are interested in seeing how you arrived at your final model. In just a sentence, please provide a reason or something that you learned from each specification.
3. **A Residuals-vs-Fitted-values Plot.** Please generate this plot for your final model that includes variable transformations. Your instructor will use this plot to assess how well you have captured the shape of the relationship between your X and your Y variable. For example, if there is a clear parabolic pattern in your residuals-vs-fitted plot, that is a signal that you should have included a square term.

Evaluation Criteria

We present the following criteria to guide you to a professional-quality report. Moreover, these criteria are also the ones we will use to grade your report. The descriptions below are copied directly from our grading rubric.

1. Intro

An introduction that is scored in the top level has very successfully made the case for the study. It will have explained why the topic is interesting, and provided compelling reasons to care about not just the general area, but rather about every concept in the research question and the statistical results to be generated. The introduction will be engaging from the very first sentence, and create a logical story that leads the reader step-by-step to the research question. After reading the introduction, no part of the research question will appear arbitrary or unexpected, instead flowing naturally from the background provided.

2. Description of the Data Source

A report that is scored in the top level will describe the provenance of the data; the audience will know the source of the data, the method used to collect the data, the units of observation of the data, and important features of the data that are useful for judging the analysis.

3. Data Wrangling

A report that is scored in the top level on data wrangling will have succeeded to produce a modern, legible data pipeline from raw data to data for analysis. Because there are many pieces of data that are being marshaled, the reports that score in the top level will have refactored different steps in the data handling into separate files, functions, or other units. It should be clear what, and how, any additional features are derived from this data. The analysis avoid defining additional data frames when a single source of truth data would suffice.

4. Operationalization

A report that is scored in the top level on operationalization will have precisely articulated and justified the decisions leading to the variables used in the analysis. The reader will be left with a clear understanding of the concepts in the research question, and how they relate to the operational definitions, including any major gaps that may impact the interpretation of results. You should also list how many observations you remove and for what reasons. When there is more than one reasonable way to operationalize a concept, the report will explain the alternatives and provide reasons for the one that was selected. Note that there is often more than one way to operationalize a concept; we are less interested in whether you make the best possible choice, and more in how well you defend your decisions.

5. One or More Data Visualization(s)

This is a task of description, and one of the most effective ways to describe relationships between variables is through visualization.

You are required to include at least one plot in your main report that highlights the relationship between your Y and your X variables In your text, you should provide your interpretation of what this plot means, and link to the plot. For example, we might write that Figure 1 displays 300 observations from a synthetic model of Earnings and Age, showing an increasing trend though the relationship is less positive for higher ages.

Include a visual representation of your final model predictions on the plot. A report that is scored in the top level will have plots that effectively transmit information, engage the reader's interest, maximize usability, and follow best practices of data visualization. Titles and labels will be informative and written in plain English, avoiding variable names or other artifacts of R code. Plots will have a good ratio of information to space or information to ink; a large or complicated plot will not be used when simple plot or table would show the same information more directly. Axis limits will be chosen to minimize visual distortion and avoid misleading the viewer. Plots will be free of visual artifacts created by binning. Colors and line types will be chosen to reinforce the meanings of variable levels and with thought given to accessibility for the visually-impaired. Your report will be free of "output dumps" - code output that has not been formatted for human-readability. Every single plot and table in the report will be discussed in the narrative of the report.

6. Model Specification

A report that is scored in the top level will have chosen a set of regression models that strongly support the goal of the study. Variables transformations will be chosen to inform the reader of the shape of the joint distribution, and will be human-understandable. A reason will be provided for the chosen variable transformations. Ordinal variables will not be treated as metric. All model specifications will be displayed in a regression table, using a package like `stargazer` to format your output. Displayed standard errors will be correctly chosen.

7. Model Assumptions

A report that scores in the top-level has provided an thorough and precise assessment of the assumptions supporting the regression. The list of assumptions is appropriate given the sample size and modeling goals. Each assumption is evaluated fairly, and discussed defensively - without overstating how credible the assumption is or rendering a final up-or-down judgement on the assumption. The report will not miss any important violations of any assumption. Where possible, the report discusses the consequences for the analysis of a violated assumption.

Table 1: Regression Results.

	Dependent variable:		
	earnings		
	(1)	(2)	(3)
Age	39.49^{***}	32.75	120.85
	(1.59)		
Age^2			−1.10
Education		91.92	45.74
Intercept		167.05	106.56
education13		214.85	145.47
education14		254.06	192.13
education15		295.85	246.05
education16		270.78	284.23
education17		247.61	311.41
education18		71.27	419.89
education19		−155.33	386.04
Constant	270.02^{***}	343.65	−1,334.24
	(63.04)		
Observations	300	300	300
R^2	0.89	0.96	0.98
Adjusted R^2	0.89	0.95	0.98
Note:	$^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$		

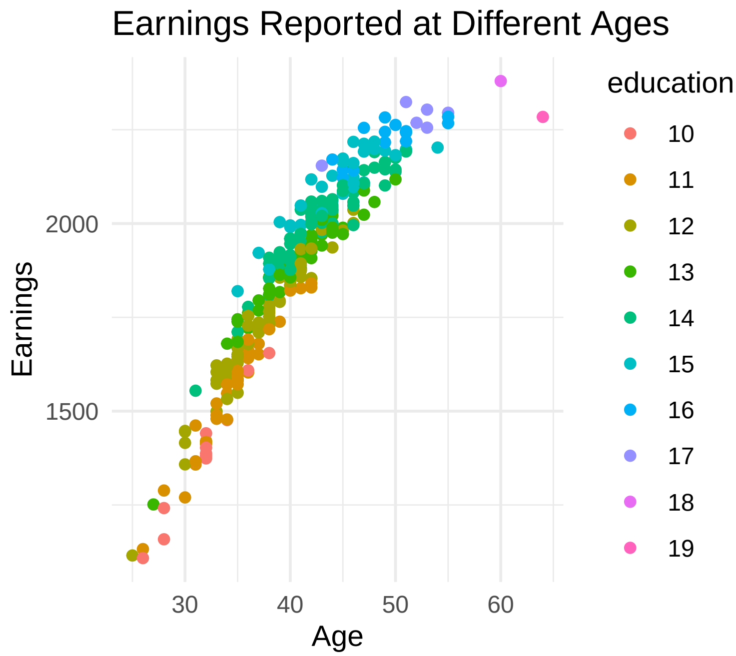


Figure 1: Synthetic data showing a hypothetical relationship between Earnings, Age, and Education. An increasing trend can be seen with decreasing slope for higher ages.

8. Model Results and Interpretation

A report that scores in the top level will correctly interpret statistical significance, clearly interpret practical significance, and comment on the broader implications of the results. It may want to include statistical tests besides the standard t-tests for regression coefficients. When discussing practical significance, comment on both the direction and magnitude of your coefficients, placing them in context so the reader can understand if they are important. To help the reader understand your fitted model, you may want to describe hypothetical datapoints (e.g. a hypothetical person with 1 cat is predicted to spend $2400 on pet care. That rises to $3200 for a hypothetical person with 2 cats…).

9. Overall Effect

A report that scores in the top level will have met expectations for professionalism in data-based writing, reasoning and argument for this point in the course. It can be presented, as is, to another student in the course, and that student could read, interpret and take away the aims, intents, and conclusions of the report.

References